

ORIGINAL RESEARCH

PERIPHERAL

2-Year Patient-Centered and Clinical Outcomes of Sirolimus- Versus Paclitaxel-Coated Balloon Angioplasty in Femoropopliteal Artery Disease



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ABSTRACT

BACKGROUND Drug-coated balloons improve outcomes after femoropopliteal angioplasty, but concerns regarding the long-term safety of paclitaxel-coated devices have prompted interest in alternative antiproliferative strategies.

OBJECTIVES This study sought to evaluate the 2-year patient-centered and clinical outcomes of sirolimus-coated balloon (SCB) versus paclitaxel-coated balloon (PCB) angioplasty.

METHODS SIRONA was a prospective, multicenter, randomized, controlled, noninferiority trial comparing SCB with PCB angioplasty in patients with femoropopliteal artery disease (96.5% intermittent claudication). The present analysis reports prespecified outcomes at 2 years. Effectiveness endpoints included patient-reported outcomes (VascuQoL, clinical improvement, and EQ-5D-3L), primary patency, and freedom from clinically driven target lesion revascularization. Safety endpoints included major amputation and all-cause mortality. Longer-term follow-up is ongoing to assess durability and safety up to 5 years.

RESULTS A total of 482 patients was randomized (SCB: n = 238; PCB: n = 244). Improvements in vascular quality of life were sustained through 2 years and were similar between groups (between-group difference: 0.05, 95% CI: -0.15 to 0.24; $P = 0.63$). Clinical improvement (≥ 1 Rutherford category) occurred in 88% of the SCB group and 89% of the PCB group ($P = 0.89$). Kaplan-Meier estimates of primary patency were 64.6% for SCB and 67.2% for PCB (log-rank $P = 0.34$). Freedom from clinically driven target lesion revascularization was 91.2% for SCB and 88.1% for PCB (log-rank $P = 0.42$). Major amputation was rare (0.4% in both groups). All-cause mortality occurred in 6.3% and 3.7%, respectively ($P = 0.21$), with no device- or procedure-related deaths.

CONCLUSIONS SCB angioplasty resulted in sustained patient-reported and clinical outcomes comparable to PCB angioplasty through 2 years, with similar patency, revascularization, and safety. (JACC Cardiovasc Interv. 2026;19:2251-2261) © 2026 Published by Elsevier on behalf of the American College of Cardiology Foundation.

ABBREVIATIONS AND ACRONYMS

cdTLR = clinically driven
target lesion revascularization

cdTVR = clinically driven
target vessel revascularization

DCB = drug-coated balloon

MID = minimal clinically
important difference

PACSS = peripheral arterial
calcium scoring system

PCB = paclitaxel-coated
balloon

SCB = sirolimus-coated
balloon

TVR = target vessel
revascularization

Endovascular revascularization is commonly performed to relieve symptoms in patients with intermittent claudication. However, restenosis after balloon angioplasty remains a major limitation. Mechanical stress within the femoropopliteal segment and the biological response to vascular injury, particularly in chronic total occlusions and calcified lesions, contribute to recurrent vessel narrowing and the need for repeat revascularization.^{1,2} The primary goals of treatment for patients with intermittent claudication extend beyond vessel patency to include improving vascular quality of life and relieving symptoms.

Drug-coated balloons (DCBs) improve the outcomes of femoropopliteal interventions by delivering an antiproliferative drug to the vessel wall without leaving a permanent implant behind. Paclitaxel-coated balloons (PCBs) have demonstrated superior patency and a reduced need for repeat revascularization in randomized trials and meta-analyses, even in the long term, compared with uncoated balloon angioplasty.^{3,4} However, concerns about a potential late mortality signal associated with paclitaxel-coated devices have sparked debate and interest in alternative antiproliferative agents for peripheral endovascular therapy.^{3,5–7}

Sirolimus is a cytostatic drug that inhibits smooth muscle cell proliferation and migration via the mammalian target of rapamycin (mTOR) pathway. Advances in coating technologies have enabled the development of sirolimus-coated balloons (SCBs), which are designed to enhance sirolimus transfer and

retention within the vessel wall. Prospective single-arm studies, including the SELUTION SFA and XTOSI trials, have reported promising outcomes through up to 3 years following SCB angioplasty for femoropopliteal artery disease.^{8–10} Additionally, the randomized SIRONA trial demonstrated that, at 12 months, SCB angioplasty is noninferior to PCB angioplasty in terms of primary patency and the composite clinical endpoint of freedom from target vessel revascularization, major amputation, and device- or procedure-related death.¹¹ Improvements in vascular quality of life and clinical symptoms were also comparable. These results imply that SCB could be a valuable alternative to PCB. Here, we report the prespecified 2-year outcomes of the SIRONA trial, with a particular emphasis on the durability of patient-reported outcomes through 2 years, including vascular quality of life and clinical improvement.

METHODS

STUDY DESIGN AND PATIENT POPULATION. The SIRONA trial is a prospective, multicenter, randomized, controlled, noninferiority trial comparing SCB angioplasty with PCB angioplasty for the treatment of femoropopliteal artery disease. The design and 12-month results have been reported previously.^{11,12}

Patients with peripheral artery disease corresponding to Rutherford categories 2 to 4 were eligible. Target lesions were required to be single lesions located in the superficial femoral artery and/or proximal popliteal artery. Successful intraluminal guide-wire crossing and predilation without flow-limiting

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

dissection were required before randomization, as was the presence of at least 1 patent infrapopliteal runoff vessel ($<50\%$ stenosis). Key exclusion criteria were in-stent restenosis and severe calcification. After confirmation of eligibility, patients were randomized in a 1:1 ratio to SCB or PCB angioplasty groups using a computer-generated sequence with block randomization stratified by center.

The study was conducted at 25 centers in Germany and Austria in accordance with the Declaration of Helsinki and ISO 14155 standards. The protocol was approved by the institutional ethics committee at each participating center, and all patients provided written informed consent. Duplex ultrasound and angiographic data were analyzed by a blinded core laboratory (CoreLab Black Forest GmbH). Study data were monitored by a contract research organization (VascuScience GmbH). An external data safety monitoring board oversaw safety, and an external, blinded clinical events committee adjudicated the primary outcomes. Patients, core laboratory personnel, and members of the clinical events committee were blinded to treatment assignment, whereas investigators, data safety monitoring board members, and treating physicians were not. SIRONA has been prospectively registered with ClinicalTrials.gov ([NCT04475783](https://clinicaltrials.gov/ct2/show/study/NCT04475783)) and the German Clinical Trials Register ([DRKS00022452](https://www.drks.de/DRKS00022452)). The present analysis reports the prespecified outcomes at 2 years.

STUDY DEVICE AND INTERVENTION. Endovascular procedures were performed according to contemporary guidelines and best clinical practice. Predilation of the target lesion was recommended for at least 60 seconds. Patients randomized to the intervention arm (SCB group) were treated with the Magic Touch PTA sirolimus-coated balloon (Concept Medical Inc), a polymer-free device with phospholipid-based sirolimus nanocarriers with an active substance concentration on the surface of $1.27 \mu\text{g}/\text{mm}^2$. Patients assigned to the control arm (PCB group) were treated with commercially available paclitaxel-coated balloons with established efficacy for femoropopliteal disease, used according to manufacturer instructions.

Balloon diameter was selected to match the reference vessel diameter, and balloon length was chosen to cover the lesion with at least 1-cm margins proximally and distally. Balloon inflation was maintained for at least 60 seconds, with longer inflation encouraged. If more than one device was required, balloons were overlapped to ensure full lesion coverage. Adjunctive therapy was permitted at the operator's discretion. Postdilation was allowed for residual stenosis $\geq 30\%$. Bailout stenting with bare

nitinol stents was required for persistent residual stenosis $\geq 50\%$ or flow-limiting dissection. Periprocedural antithrombotic therapy followed guideline-recommended practice. Dual antiplatelet therapy was recommended before the procedure or administered as a loading dose and continued for at least 4 weeks after the intervention, followed by long-term single antiplatelet therapy.

ENDPOINTS. Primary 12-month endpoints of the SIRONA trial have been reported previously.¹¹ The present analysis focuses on prespecified effectiveness and safety outcomes at 2 years (790 days after randomization). Effectiveness endpoints were patient-reported outcome measures as well as clinical and hemodynamic outcomes. Patient-reported outcome measures included the vascular quality of life, assessed by the 25-item VascuQoL questionnaire¹³ and the health-related quality of life using the European Quality of Life-5 Dimensions-3 Level (EQ-5D-3L) instrument.¹⁴ The minimal clinically relevant difference (MID) in the vascular quality of life score is reported with 0.87 for improvement and 0.23 for deterioration.¹⁵ The MID for the health-related quality of life index in nonsurgical patients ranges from 0.04 to 0.1.¹⁶ Clinical effectiveness outcomes included clinical improvement, defined as an improvement of ≥ 1 Rutherford category; pain-free and maximum walking distance, based on the 6-minute walk test, without the need for target vessel revascularization (TVR); as well as freedom from clinically driven target lesion revascularization (cdTLR) and clinically driven TVR (cdTVR). Hemodynamic effectiveness outcomes included primary patency, defined as freedom from duplex ultrasound-defined restenosis (peak systolic velocity ratio: >2.4), secondary patency, defined as patency maintained with or without repeat endovascular intervention, and hemodynamic improvement (increase in ankle-brachial index ≥ 0.15 or to ≥ 0.9 without the need for TVR). Safety endpoints included amputation-free survival, major target-limb amputation, and all-cause mortality.

STATISTICAL ANALYSIS. Statistical analyses were performed according to the available-case intention-to-treat principle. Continuous variables are presented as mean $\pm$ SD, and categorical variables are presented as counts with percentages. Between-group comparisons of categorical variables were performed using the chi-square test or Fisher exact test, and continuous variables were compared using the Student's *t*-test or Mann-Whitney *U* test. Effect estimates are reported as mean differences, odds ratios, or risk differences with 95% confidence

TABLE 1 Participant and Core Laboratory-Reported Lesion Characteristics by Treatment Group

	SCB Group (n = 238)	PCB Group (n = 244)
Age, y	67.7 ± 9.1	68.3 ± 8.6
Male	152 (63.9)	159 (65.2)
Current smoker	102/236 (43.2)	114/241 (47.3)
Former smoker	97/236 (41.1)	94/241 (39.8)
Diabetes mellitus	81 (34.0)	81 (33.2)
Hypertension	209 (87.8)	209 (85.7)
Hyperlipidemia	181 (76.1)	195 (79.9)
Coronary artery disease	71 (29.8)	85 (34.8)
Renal insufficiency	41 (17.2)	45 (18.4)
Rutherford classification		
Category 2	58 (24.4)	65 (27.0)
Category 3	173 (72.7)	168 (68.9)
Category 4	5 (2.1)	10 (4.1)
Category 5	2 (0.8)	0 (0)
Ankle-brachial index of target limb ^a	0.65 ± 0.21	0.67 ± 0.24
Lesion length, mm	84.3 ± 61.7	84.1 ± 60.1
Total occlusion	81 (34.0)	79 (32.4)
Calcification PACSS grade 3	100/223 (44.8)	102/219 (46.6)
Calcification PACSS grade 4	64/223 (28.7)	61/219 (27.9)
Stenosis, % of vessel diameter	83.0 ± 16.1	81.7 ± 17.5
No patent runoff vessel before procedure ^b	30/178 (16.9)	25/187 (13.4)

Values are mean ± SD or n (%). There were no significant differences between the groups. ^aData for the ankle-brachial index of the target limb are shown for 227 and 235 participants of the SCB and the PCB group, respectively. ^bPatency of outflow vessels was defined as <50% diameter stenosis.

PACSS = peripheral arterial calcium scoring system; PCB = paclitaxel-coated balloon; SCB = sirolimus-coated balloon.

TABLE 2 Procedural Characteristics by Treatment Group

	SCB Group (n = 238)	PCB Group (n = 244)
Core laboratory-reported		
Dissection after treatment with DCB		
NHLBI type B	85/219 (38.8)	111/231 (48.1)
NHLBI type C	1/219 (0.5)	1/231 (0.4)
NHLBI type D	92/219 (42.0)	78/231 (33.8)
NHLBI type E	1/219 (0.5)	6/231 (2.6)
Distal embolization	4 (1.7)	3/243 (1.3)
Residual diameter stenosis, %	24.7 ± 11.6	27.6 ± 11.7
Residual stenosis >30%	74/236 (31.4)	89/242 (36.8)
Site-reported		
Adjunctive treatment performed ^a	92 (38.7)	107 (43.9)
Postdilation with uncoated balloon	75 (31.6)	83 (34.0)
Drug-coated balloon angioplasty ^b	7 (2.9)	16 (6.6)
Bailout stent implantation ^c	56 (23.5)	50 (20.5)

Values are n (%) or mean ± SD. There were no significant differences between the groups. ^aIn 133 (66.8%) of the 199 participants who underwent adjunctive treatment, a dissection had been diagnosed by the investigators. ^bAn SCB of another vendor (SELUTION SLR, Med Alliance SA) was used as adjunctive treatment in a participant of the intervention group. ^cIn 58 (54.7%) of the 106 participants who underwent stent implantation, the treatment was performed because of a dissection.

DCB = drug-coated balloon; NHLBI = National Heart, Lung, and Blood Institute grading system; PCB = paclitaxel-coated balloon; SCB = sirolimus-coated balloon.

intervals. Time-to-event outcomes, including primary and secondary patency, freedom from cdTLR, and amputation-free survival, were analyzed using Kaplan-Meier methods and compared using log-rank tests. Hazard ratios with 95% confidence intervals were estimated using Cox proportional hazards models.

All statistical tests were 2-sided, and a *P* value of <0.05 was considered statistically significant. Analyses were performed using SAS software (version 9.4; SAS Institute) and XLSTAT (version 2015.6.01.24026; Addinsoft).

RESULTS

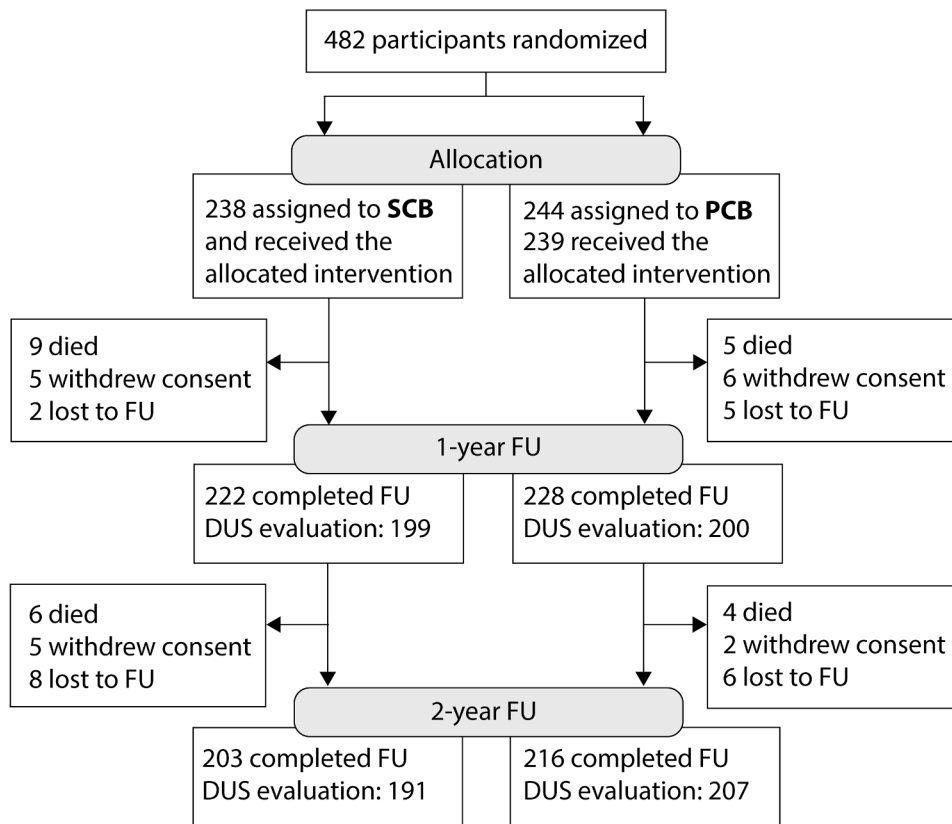
STUDY POPULATION. From April 21, 2021, to September 23, 2022, a total of 482 patients were randomized to the SCB group (n = 238) or PCB group (n = 244). Baseline clinical and lesion characteristics were well balanced between the groups and have been reported previously.¹¹ Key characteristics are summarized in **Tables 1 and 2**. Most participants presented with intermittent claudication (96.3%). The mean lesion length was 8.4 ± 6.1 cm, and 33.2% of the lesions were chronic total occlusions. Severe calcification (peripheral arterial calcium scoring system [PACSS] grade 4) was present in 28.3% of the lesions. Bailout stenting was performed in 22.0% of the procedures. Residual stenosis >30% was present in 31.4% participants in the SCB group and in 36.8% of participants in the PCB group. Flow-limiting dissection occurred in 42.5% and 36.4%, respectively. At 2 years, 85.3% of participants in the SCB group and 88.5% in the PCB group completed follow-up. Duplex ultrasound data were available for 80.3% and 84.8% of the participants, respectively (**Figure 1**).

PATIENT-REPORTED OUTCOMES AND CLINICAL EFFECTIVENESS

Vascular quality of life as assessed by the 25-item VasculQoL questionnaire improved by at least the MID in 70% of the SCB group and in 66% of the PCB group (*P* = 0.49). The baseline-adjusted between-group difference in the VasculQoL score was 0.05 (95% CI: 0.15-0.24; *P* = 0.63) after 2 years (**Central Illustration**). In both groups, improvement was strongest in the Pain and Activities domains (**Table 3**). Health-related quality of life as measured by the EQ-5D-3L index improved by ≥0.1 (upper limit of the MID range) in 66% (116/176) of participants in the SCB group and 63% (114/180) of participants in the PCB group (*P* = 0.69).

Clinical improvement of ≥1 Rutherford category was achieved in 88% of the SCB group and 89% of

FIGURE 1 Patient Flow Diagram Through 2 Years



Two-year available case intention-to-treat analyses included all randomized participants with data on the secondary outcomes. The deaths were not device- or procedure-related. DUS = duplex ultrasound; FU = follow-up; PCB = paclitaxel-coated balloon; SCB = sirolimus-coated balloon.

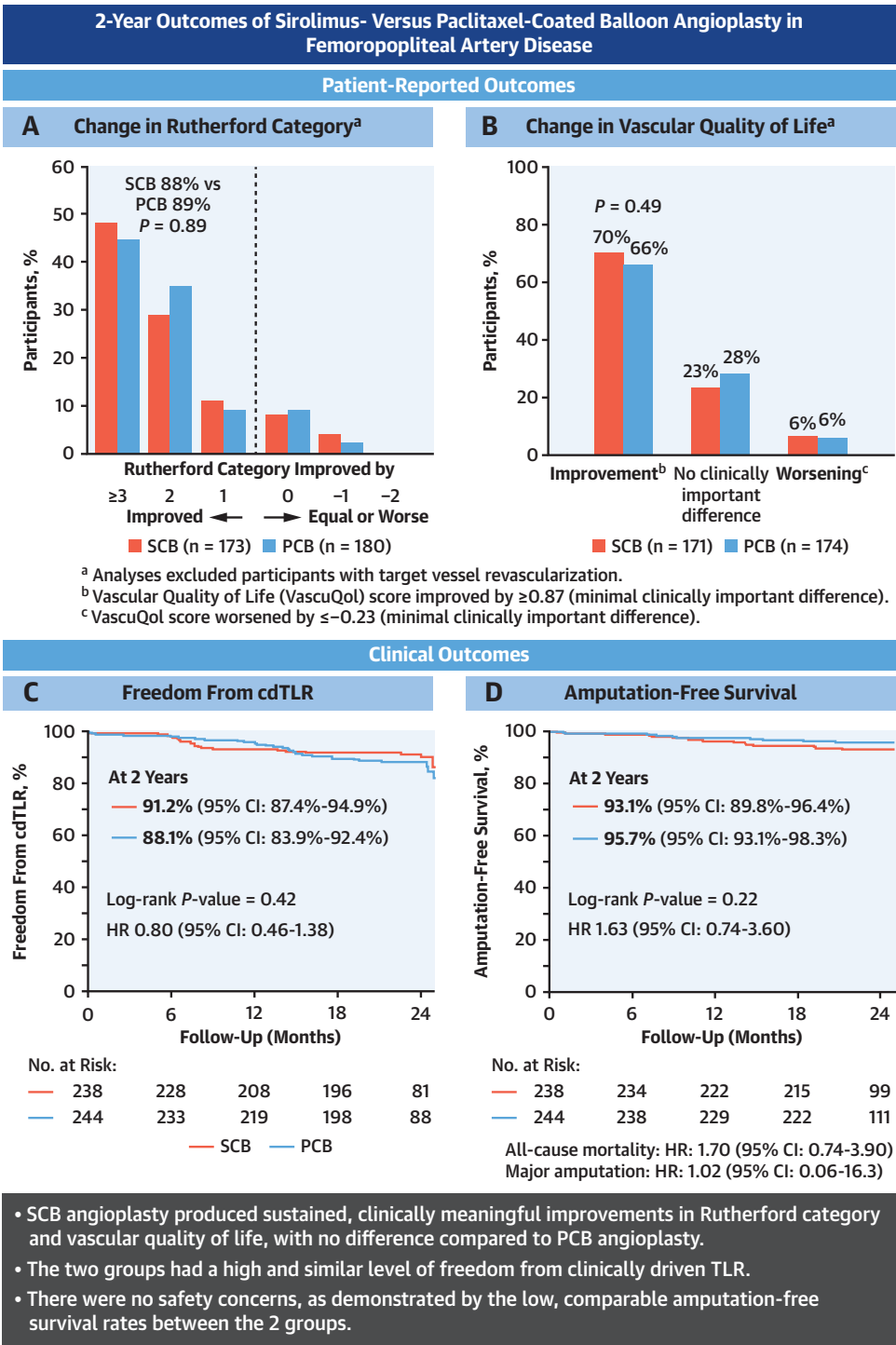
the PCB group ($P = 0.89$) (Central Illustration, Table 4). The magnitude of improvement did not differ between groups (OR for greater improvement with SCB vs PCB: 1.0 [95% CI: 0.6-1.5]; $P = 0.82$) (Table 3). The Rutherford category at baseline did not interact with the SCB vs PCB treatment effect regarding improvement in VascuQol scores or Rutherford category at 2 years (Supplemental Tables 1 and 2).

Pain-free walking distance, maximum walking distance, and health-related quality of life improved in both groups, without significant between-group differences (Table 3). Freedom from cdTLR at 2 years was 91.2% in the SCB group and 88.1% in the PCB group (log-rank $P = 0.42$) (Central Illustration). The cdTLR rate after PCB varied depending on the type of PCB used as the control. However, the SCB did not differ significantly from any of the 4 main PCB types (Supplemental Figure 1).

HEMODYNAMIC OUTCOMES. Hemodynamic improvement, defined as an increase in the ankle-brachial index by ≥ 0.15 or to ≥ 0.9 , occurred in 74.2% of participants in the SCB group and 69.0% participants in the PCB group at 2 years ($P = 0.36$). Binary restenosis rates were similar between groups (SCB: 38.2% vs PCB: 33.8%, $P = 0.36$) (Table 4). Kaplan-Meier estimates of primary patency were 64.6% for SCB and 67.2% for PCB (log-rank $P = 0.34$), while secondary patency was 71.9% and 81.7%, respectively (log-rank $P = 0.052$) (Figure 2). Although the primary patency rate varied depending on the type of PCB used as the control, the SCB did not differ significantly from any of the 4 main PCB types (Supplemental Figure 2).

SAFETY OUTCOMES. Amputation-free survival at 2 years did not differ between groups (SCB: 93.1% vs PCB: 95.7%, log-rank $P = 0.22$; HR: 1.63 [95% CI: 0.74-3.60]). The hazard ratio was primarily influenced by all-cause mortality (SCB vs PCB, HR: 1.70

CENTRAL ILLUSTRATION Patient-Reported and Clinical Outcomes Through 2 Years



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(A and B) changes from baseline to 2 years in Rutherford category (A) and vascular quality of life (B). (C and D) Two-year survival curves for freedom from clinically driven target lesion revascularization (C) and amputation-free survival (D). cdTLR = clinically driven target lesion revascularization; PCB = paclitaxel-coated balloon; SCB = sirolimus-coated balloon.

TABLE 3 Functional and Hemodynamic Improvement at 2 Years^a

	SCB Group (n = 238)		PCB Group (n = 244)		Between-Group Difference ^b (95% CI)	P Value
	No. Analyzed	Improvement	No. Analyzed	Improvement		
Rutherford classification, median (IQR) ^c	173	2.0 (2.0-3.0)	180	2.0 (2.0-3.0)	OR: 1.0 (0.6 to 1.5)	0.82
Resting ankle-brachial index	163	0.25 ± 0.23	168	0.24 ± 0.26	0.01 (−0.04 to 0.06)	0.82
Pain-free walking distance, m ^d	153	197 ± 170	150	198 ± 167	1 (−39 to 37)	0.96
Maximum walking distance, m ^d	153	110 ± 137	150	100 ± 148	10 (−22 to 42)	0.55
VascuQoL score ^e	171	1.4 ± 1.1	174	1.3 ± 1.1	0.09 (−0.14 to 0.32)	0.45
Pain	176	1.7 ± 1.6	180	1.6 ± 1.4	0.14 (−0.17 to 0.46)	0.39
Symptoms	176	1.0 ± 1.1	179	1.1 ± 1.2	0.04 (−0.2 to 0.28)	0.78
Activities	173	1.7 ± 1.2	180	1.6 ± 1.3	0.14 (−0.12 to 0.40)	0.30
Social	176	1.4 ± 1.8	179	1.4 ± 1.9	−0.05 (−0.43 to 0.35)	0.81
Emotions	175	1.2 ± 1.4	179	1.2 ± 1.4	0.06 (−0.23 to 0.35)	0.70
Health-related quality of life, EQ-5D-3L index ^f	176	0.2 ± 0.3	180	0.1 ± 0.3	0.03 (−0.03 to 0.09)	0.36

Values are mean ± SD unless otherwise indicated. Analysis referred to available case intention-to-treat analysis. ^aParticipants who underwent TVR after the index procedure were excluded from the analyses. ^bThe measure of effect is reported as the mean difference between groups or, in the case of the Rutherford category, as the odds ratio assessed by ordinal regression. ^cClassification ranges from categories 1 to 6. As the category approaches 6, the clinical condition worsens. ^dWalking distance was assessed by the 6-minute walk test. ^eThe score is derived from the 25-item Vascular Quality of Life (VascuQoL) questionnaire. The domains encompass pain, symptoms, activities, social life, and emotional well-being. The score ranges from 1 to 7 (worse to best vascular quality of life). ^fThe index was calculated using the European Quality of Life-5 Dimensions-3 Level (EQ-5D-3L) instrument. Index ranges from 0 to 1. As the index approaches 1, the assessed quality of life increases.

PCB = paclitaxel-coated balloon; SCB = sirolimus-coated balloon.

(95% CI: 0.74-3.90) (**Central Illustration**). One participant per group underwent major amputation at 1 and 7 months. All-cause death occurred in 6.3% of SCB participants and 3.7% of PCB participants ($P = 0.21$) (**Table 4**). Causes of death were similar between the groups (**Table 5**). The clinical events committee did not consider any death to be device- or procedure-related.

DISCUSSION

The present head-to-head comparison from the SIRONA trial builds on the previously reported

12-month results¹¹ and provides randomized evidence on the durability of treatment effects through 2 years after SCB angioplasty for femoropopliteal artery disease. Patients experienced sustained improvements in both patient-reported and clinical outcomes, with clinically meaningful gains in vascular and health-related quality of life comparable to those observed with PCB angioplasty. These benefits were accompanied by durable symptomatic relief, with nearly 90% of patients achieving an improvement of ≥ 1 Rutherford category. Freedom from cdTLR and primary patency remained similar between treatment strategies through 2 years. Major

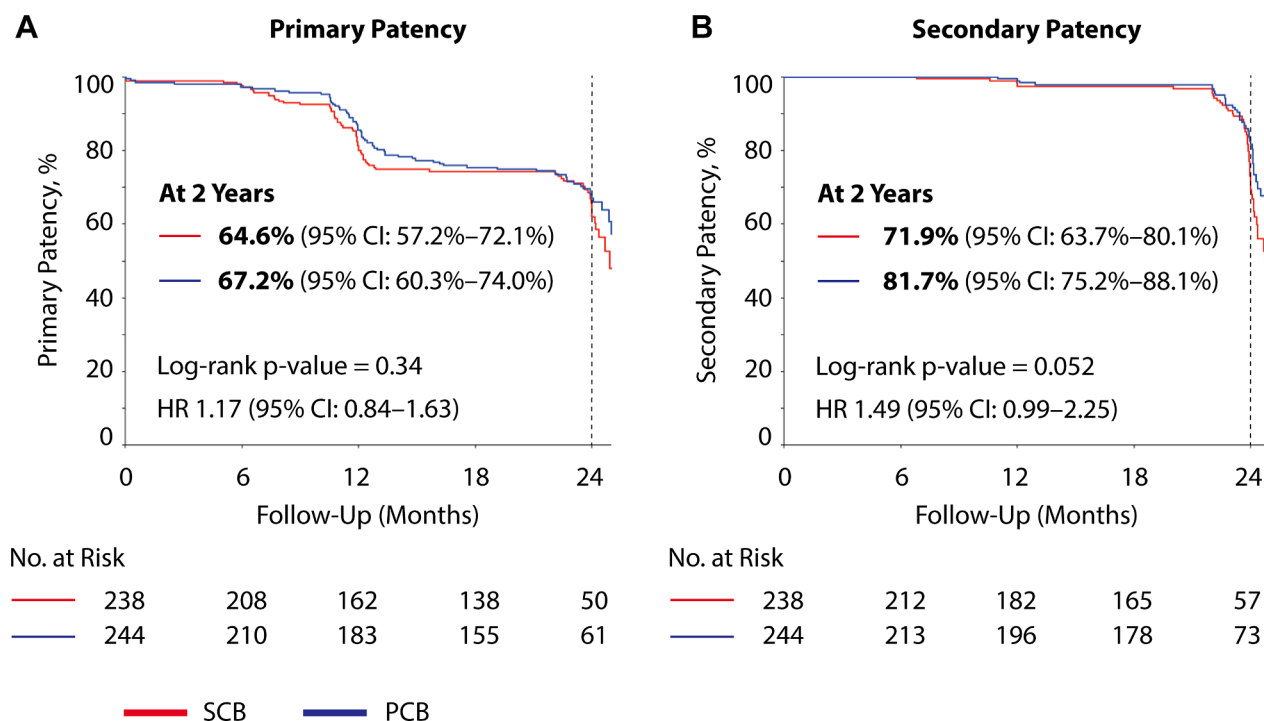
TABLE 4 Effectiveness and Safety Outcomes at 2 Years

	SCB Group (n = 238)	PCB Group (n = 244)	Risk Difference (95% CI) ^a	P Value
Clinical improvement ^{b,c}	152/173 (87.9)	160/180 (88.9)	−1.0 (−7.7 to 5.7)	0.89
Hemodynamic improvement ^{b,d}	121/163 (74.2)	116/168 (69.0)	5.2 (−4.5 to 14.9)	0.36
Binary restenosis ^e	73/191 (38.2)	70/207 (33.8)	4.4 (−5.0 to 13.9)	0.36
Clinically driven TVR ^f	32/209 (15.3)	34/247 (13.8)	1.5 (−4.9 to 8.1)	0.74
Clinically driven TLR	23/206 (11.2)	31/183 (14.5)	−3.3 (−9.7 to 3.1)	0.38
Major target-limb amputation ^{f,g}	1/238 (0.4)	1/244 (0.4)	0.0 (−1.1 to 1.2)	0.99
All-cause death ^f	15/238 (6.3)	9/244 (3.7)	2.6 (−1.2 to 6.6)	0.21
Device- or procedure-related death	0/238 (0)	0/244 (0)	—	—

Values are n/N (%). Analysis referred to available case intention-to-treat analysis. ^aDifferences are shown in percentage points. ^bParticipants who underwent TVR after the index procedure were excluded from the analyses. ^cImprovement by ≥ 1 Rutherford category. ^dParticipants who underwent TVR were excluded from the analysis. ^eImprovement in resting ankle-brachial index by ≥ 0.15 or to ≥ 0.9 . ^fDuplex-defined binary restenosis (peak systolic velocity ratio >2.4) as adjudicated by the core laboratory. ^gEvent data on clinically driven TVR, major amputation, and death were included in the analysis up to 790 days after randomization. ^hMajor amputation was conducted at 7 months (SCB) and 1 month (PCB). It was preceded by clinically driven target lesion revascularization.

PCB = paclitaxel-coated balloon; RD = risk difference; SCB = sirolimus-coated balloon; TLR = target lesion revascularization; TVR = target vessel revascularization.

FIGURE 2 Kaplan-Meier Curves of Primary and Secondary Patency Through 2 Years



Two-year survival curves for (A) primary patency and (B) secondary patency. Primary patency was defined as the absence of target lesion restenosis as assessed by duplex ultrasound and indicated by a peak systolic velocity ratio of >2.4 or total occlusion, as adjudicated by the masked core laboratory, without the need for additional target lesion revascularization. Secondary patency was defined as patency obtained with or without the use of an additional endovascular procedure. Core laboratory-adjudicated duplex ultrasound evaluation of target lesion patency was scheduled at 12 and 24 months or in the event of target lesion revascularization. PCB = paclitaxel-coated balloon; SCB = sirolimus-coated balloon.

amputation and all-cause mortality were rare, and although event numbers were low, no significant imbalance between groups was observed.

At 2 years, improvements in vascular quality of life were sustained, with no significant difference between groups. The proportion of participants

showing clinically relevant improvement¹⁵ remained stable over time. The greatest effects continue to be seen in the Pain and Activities domains of the VasculQoL questionnaire in both groups. This likely reflects that these domains are closely aligned with the symptom burden of intermittent claudication. In contrast, the Symptoms and Emotions domains also include items concerning critical limb-threatening ischemia.¹⁷ Improvements in health-related quality of life were maintained in both groups. Although the improvement in the maximum walking distance was slightly smaller in both groups at the 2-year follow-up, improvements in maximum and pain-free walking distance and Rutherford categories over time supported the durability of patient-centered benefits beyond 1 year.

The SIRONA study results on patient-centered outcomes confirmed the effectiveness of SCB angioplasty with regard to patient-centered outcomes previously described in the SELUTION SFA Japan single-arm trial,⁹ which reported results through up

TABLE 5 Causes of Death at 2 Years

	SCB Group (n = 238)	PCB Group (n = 244)	P Value
Infectious diseases	5 (2.1)	3 (1.2)	0.50
Cardiovascular diseases	5 (2.1)	3 (1.2)	0.50
Neoplasms	5 (2.1)	1 (0.4)	0.12
Endocrine diseases	0 (0)	1 (0.4)	>0.99
Unknown	0 (0)	1 (0.4)	>0.99

Values are n (%). All deaths were adjudicated by the data safety and monitoring board as not related to the study device or procedure. Available case intention-to-treat analysis through 790 days. PCB = paclitaxel-coated balloon; SCB = sirolimus-coated balloon.

to 3 years. The SELUTION SFA Japan trial revealed that a high proportion of participants (81%) were asymptomatic at 2 years. However, in contrast to the SIRONA trial, the SELUTION SFA Japan trial probably included participants who underwent TVR in the analysis. Furthermore, the Rutherford categories of the SIRONA trial participants were slightly worse at baseline.

Evidence on patient-centered outcomes beyond 1 year after PCB angioplasty remains limited.^{3,4} Prior randomized trials comparing PCB with uncoated balloon angioplasty have shown inconsistent effects on symptoms and quality of life.¹⁸⁻²⁰ The more recent SWEDEPAD 2 study²¹ on paclitaxel-coated devices suggest that improvements in patient-reported outcomes may not be solely attributable to drug delivery.

However, previous meta-analyses on PCBs have demonstrated hemodynamic and clinical efficacy beyond 2 years.^{3,4} Likelihood estimates for primary patency and cdTLR favored PCB compared to standard balloon angioplasty (10 randomized controlled trials; OR: 2.47 [95% CI: 1.93-3.16], *P* for heterogeneity [P_{het}] = 0.42) and OR: 0.42 [95% CI: 0.29-0.60], P_{het} = 0.08, respectively).⁴ PCB angioplasty produced 2-year primary patency rates of 46% to 90% and freedom from cdTLR of 64% to 97%.^{3,4,20} In the SIRONA trial, the primary patency rates were similar for SCB and PCB angioplasty at 2 years, consistent with the 12-month findings.¹¹ However, the secondary patency rates were higher for the PCB group than the SCB group, though the difference was not significant. This difference was reflected in the cdTLR rates, which fluctuated throughout the 2-year period. For up to 1 year, the risk of cdTLR was slightly higher with SCB than with PCB, however, after 1 year, the risk decreased. However, these observations remain exploratory and require confirmation with longer term follow-up. It remains to be seen whether the PCB group will have a cdTLR-deferring but not cdTLR-preventing effect in the long term, as was previously observed in the SWEDEPAD 2 study.²¹

The 2-year hemodynamic and clinical results of the SIRONA trial are in the line with previously reported results from single-arm studies on SCB angioplasty.⁸⁻¹⁰ The favorable primary patency rate of 83.2% after 2 years in the SELUTION SFA Japan trial⁹ may be because of a lower percentage of diameter stenosis, fewer chronic total occlusions, and fewer moderate-to-severe calcifications. The favorable outcome may also be attributable to the less rigorous characterization of stenosis (peak systolic velocity ratio: ≥ 2.5). Nevertheless, the freedom from cdTLR (95.4%) was comparable to that of the

SIRONA SCB group. Results from the XTOSI single-arm trial^{8,10} on SCB angioplasty are available for up to 3 years. Of the 20 patients who underwent femoropopliteal SCB angioplasty, primary patency was 53% at 2 years. However, 90% of the patients who underwent femoropopliteal angioplasty had critical limb-threatening ischemia, and their lesions were considerably longer, at 28 cm. Finally, 2-year freedom from cdTLR was 93.8%, comparable to the SIRONA SCB group and the SELUTION SFA Japan trial.

As expected, in the SIRONA trial, major amputations were rare among patients with intermittent claudication throughout the study. After one event per group during the 12-month follow-up period, no additional major amputation occurred. Although numerically more deaths occurred in the SCB group, the difference was not statistically significant, and no deaths were adjudicated as device- or procedure-related. However, the SIRONA trial did not have the power to detect differences in mortality, so these findings should be interpreted with caution.

The mortality rate in the SELUTION SFA Japan trial for SCB angioplasty was slightly lower than that in the SIRONA trial SCB group (4.5% all-cause mortality at 2 years).⁹ However, mortality will continue to be evaluated through 5 years during the SIRONA trial. The findings will then be considered alongside those of other studies, including the recent SWEDEPAD 2 study,²¹ which reported an increased risk of all-cause mortality within 5 years with paclitaxel-coated devices compared to uncoated balloon angioplasty (HR: 1.47, *P* = 0.01) in patients with intermittent claudication. Through 2 to 5 years, those investigators observed a divergence in mortality rates between the 2 groups, confirming earlier meta-analysis reports of a late safety signal with PCBs.^{3,5-7} Conversely, Koekerling et al⁴ reported a similar long-term all-cause mortality risk with PCBs versus uncoated balloon angioplasty (meta-analysis of 12 randomized controlled trials; OR: 0.96 [95% CI: 0.67-1.39], P_{het} = 0.62). In this context, the present findings from the SIRONA trial do not indicate a signal for excess mortality with either treatment strategy at 2 years but should be interpreted in light of ongoing long-term follow-up.

STUDY LIMITATIONS. Several limitations should be considered when interpreting the results of the SIRONA trial at 2 years. First, although the trial was randomized and multicenter, it was not powered to detect differences in infrequent clinical events such as mortality or major amputation at 2 years. Second, different PCB platforms were allowed in the control arm, reflecting contemporary clinical practice but

potentially introducing heterogeneity in device performance. Finally, longer term follow-up is needed to further assess the durability of treatment effects and potential late safety signals associated with DCB technologies. Continued follow-up of the SIRONA cohort through 5 years is ongoing.

CONCLUSIONS

In this randomized, multicenter trial of patients with predominantly intermittent claudication, SCB angioplasty resulted in sustained improvements in patient-reported and clinical outcomes through 2 years that were comparable to those observed with PCBs. Primary patency and freedom from cdTLR were similar between treatment strategies. No device- or procedure-related safety concerns were identified. Ongoing follow-up of the SIRONA cohort will further clarify the durability and long-term safety of this treatment approach beyond 2 years.

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institution. Prof Pech has reported participation on the clinical events committee of the SIRONA trial, and Prof Katsanos has reported participation on the data safety monitoring board of the SIRONA trial. Dr Lehmann has reported no relationships relevant to the contents of this paper to disclose.

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PERSPECTIVES

WHAT IS KNOWN? PCBs improve patency after femoropopliteal angioplasty, but concerns regarding long-term safety persist. Early randomized data, including 12-month results from the SIRONA trial, suggest that SCBs provide comparable outcomes through 1 year, while durability and patient-centered benefits beyond this time point remain uncertain.

WHAT IS NEW? In this randomized trial, SCB angioplasty provided sustained improvements in patient-reported outcomes, including quality of life, with similar patency, cdTLR, and safety compared with PCBs through 2 years.

WHAT IS NEXT? These findings suggest that SCB angioplasty is a clinically relevant alternative for patients with intermittent claudication when patient-centered outcomes are prioritized. However, longer-term follow-up and additional comparative studies are needed to confirm durability and inform treatment selection in routine practice.

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KEY WORDS angioplasty balloon, intermittent claudication, peripheral artery disease, paclitaxel, sirolimus

APPENDIX For supplemental tables and figures, please see the online version of this paper.